

Interpretable Sentinel-2 Turbidity Mapping with Ensemble Symbolic Regression

A Reproducible Visual-Programming Pipeline for the Lower Seine

Nicolas Priniotakis^{*1}

¹UniLaSalle

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Abstract

We present a **proof-of-concept** pipeline for estimating water turbidity (NTU) from Sentinel-2 surface reflectance using an ensemble of genetic-programming (GP) symbolic regressors. The pipeline is built as a visual node graph inside the open-source environment **VNStudio**, exposing every parameter on a connectable node and serializing the entire experiment to a single JSON file. Training uses 563 satellite-station matchups assembled from the French Hub'Eau Naiades network for the lower Seine basin (2017–2025); stratified subsampling of the Clear (<5 NTU) class is required to prevent the GP search from collapsing to a near-constant predictor. Ten independent GP runs aggregate into a mean turbidity map (μ) and a per-pixel ensemble-spread map (σ). Applied to a 10 m July 2023 acquisition over the Rouen reach, the rebalanced model recovers a positive predicted-vs-observed slope (0.26, vs 0.02 without subsampling) and produces WFD turbidity-class proportions consistent with low-flow summer conditions. The best-of-ensemble symbolic expression spontaneously recombines a Blue/Red ratio with an NIR/SWIR contrast — the structural skeleton of published bio-optical models for turbid water — without those features being imposed, supporting their physical relevance. We position the work as a methodological demonstration of *visual programming as a reproducibility primitive* for remote-sensing experimentation, and outline concrete next steps (sample weighting, larger spatial transfer, calibrated uncertainty) toward an operational version.

Keywords: turbidity, Sentinel-2, symbolic regression, gplearn, ensemble uncertainty, Naiades, Seine, visual programming.

1 Introduction

Operational satellite mapping of inland-water turbidity relies overwhelmingly on either fixed empirical formulas [4, 2] or end-to-end machine-learning regressors [5, 6]. The former are interpretable but inflexible; the latter are flexible but opaque. Genetic-programming symbolic regression [3, 7] sits between the two: it searches a space of closed-form expressions, returning a model that a hydrologist can inspect, criticize, and connect to physical reasoning.

We pair GP with two practical ingredients that the literature usually treats separately. First, an *ensemble* of independently seeded runs whose spread quantifies model uncertainty per pixel. Second, a *visual-programming* substrate (VNStudio) in which the entire workflow — API call, atmospheric correction, sampling, feature engineering, training, inference, raster reconstruction, statistics — is a single node graph. The visual layer is not cosmetic: it locks each preprocessing

^{*}Corresponding author: nicolas.priniotakis@unilasalle.fr

decision to a parameter on a node, and exports the whole experiment as a JSON file that can be re-opened and re-executed.

Our contribution is four-fold:

1. A reproducible Sentinel-2 turbidity pipeline whose entire state (parameters, connections, in-situ source) lives in a single 14-node graph (Fig. 1).
2. Ensemble symbolic regression with explicit per-pixel uncertainty, applied to the lower Seine over July 2023.
3. Empirical evidence that an unconstrained GP search rediscovers the Blue/Red and NIR/SWIR structural building blocks of established turbidity formulas, supporting their physical relevance.
4. A concrete diagnosis of *class-imbalance collapse* in symbolic regression on long-tailed environmental targets, with a stratified-subsampling remedy implemented inline in the graph.

This work is positioned as a **proof-of-concept**: the model’s quantitative skill remains modest (slope 0.26, predictions saturating well below the upper Nāïades range), and we explicitly do not claim operational readiness. Rather, the contribution is to expose a reproducible substrate on which improvements — sample weighting, larger geographic transfer, calibrated uncertainty — can be tested in isolation without rewriting the rest of the pipeline (§7).

2 Study area and data

2.1 Geographic scope

We focus on the lower Seine basin from the Île-de-France down to the estuary near Rouen. The *training* bounding box is wide — $[-0.0225, 48.5583, 2.8878, 49.6460]$ (WGS84, lon/lat) — to maximize Nāïades station coverage. The *inference* tile, chosen for visual detail, is a 34 km-wide window over Rouen: $[0.9522, 49.2499, 1.2621, 49.4734]$, resolved at 10 m.

2.2 Satellite data

Sentinel-2 L2A scenes were retrieved through the Copernicus Data Space catalogue (`geo_copernicus` node). Acquisition window: 2023-07-01 to 2023-07-07; the lowest-cloud scene (2023-07-04) was selected automatically. Six surface-reflectance bands were extracted: B02 (Blue, 492 nm), B03 (Green, 560 nm), B04 (Red, 665 nm), B08 (NIR, 833 nm), B11 (SWIR1, 1614 nm) and B12 (SWIR2, 2202 nm). Maximum scene-level cloud cover was capped at 10 %.

2.3 In-situ measurements

Turbidity ground truth comes from the French Hub’Eau Nāïades API (`/api/v2/qualite_rivieres/analyse_p` parameter code 1295, Formazine NTU). To overcome the per-call ceiling of 20 000 rows, the `geo_naiades` node chunks queries by year over the 2017–2025 window. Only validated measurements (`code_remarque=1`) are retained. The raw download returned 563 station-day matchups falling within ± 30 d of the satellite acquisition.

3 Methods

3.1 Pipeline overview

The pipeline (Fig. 1) splits into a *training* branch (top) and an *inference* branch (bottom), sharing a single GP ensemble. Sections 3.2–3.9 document each component and its parameters.

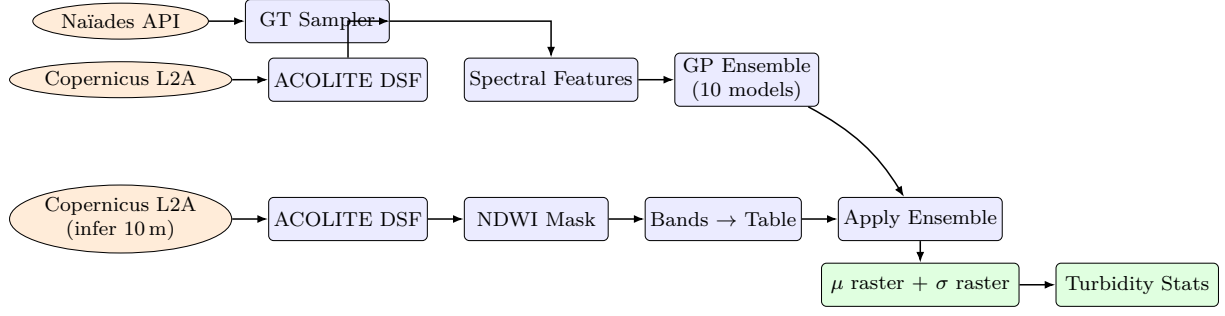


Figure 1: Pipeline graph as instantiated in VNStudio (schematic). Each box is a node; each arrow a typed connection (geotiff, dataframe, mask). The full graph is shipped as `public/templates/Machine Learning Course/gp_ensemble_uncertainty.vn`.

3.2 Atmospheric correction (`geo_acolite_full`, DOS-1 mode)

Raw L2A reflectance still carries residual aerosol scattering and surface effects. We apply a fast Dark-Object Subtraction (DOS-1) variant on the L2A stream, configured with `band_names = Blue, Green, Red, NIR, SWIR1, SWIR2` and `dsf_aot_estimate = dark_spectrum`. Two implementation choices are non-obvious:

- **Auto-scaling:** the test for “raw DN vs. reflectance” uses the 99th-percentile rather than the maximum, which would be dragged by a single saturated pixel and divide the entire scene by 10 000.
- **Dark-object selection:** the dark value is the 0.1th percentile of positive pixels, capped to discard cases where the 1st percentile coincides with the water signal. When $\text{dark} > 0.5 \times \text{median}$ the subtraction is skipped.

Output is $Rrs \in [0, 0.2] \text{ sr}^{-1}$, directly comparable to in-situ Rrs .

3.3 Ground-truth sampling (`geo_ground_truth_sampler`)

The Naïades DataFrame is joined to the satellite raster through lat/lon $\rightarrow$ pixel mapping. Each station is sampled via a 5×5 `nanmean` window (smoothing sensor noise and co-registration error) and constrained by a ± 30 d temporal window around the satellite date. Parameters:

Parameter	Value
<code>lat_col / lon_col</code>	<code>lat / lon</code>
<code>target_col</code>	<code>label (NTU)</code>
<code>band_names</code>	<code>Blue, Green, Red, NIR, SWIR1, SWIR2</code>
<code>target_min / target_max</code>	<code>0 / 500 NTU</code>
<code>mask_radius</code>	<code>3 px</code>
<code>sample_window</code>	<code>5 px (5×5 nanmean)</code>
<code>image_date</code>	<code>2023-07-04</code>
<code>date_window</code>	<code>30 d</code>

3.4 Spectral feature engineering (`ml_spectral_features`)

Identical on the training and inference branches:

- Ratios: `Red/Green`, `Blue/Green`, `Red/NIR`, `NIR/SWIR1`, `Red/SWIR1`
- Indices: $\text{NDTI} = (R - G)/(R + G)$, $\text{MNDWI} = (G - \text{SWIR1})/(G + \text{SWIR1})$
- Log transforms: `log(Red)`, `log(SWIR1)`

This expanded basis enlarges the symbolic search space without forcing a particular model shape.

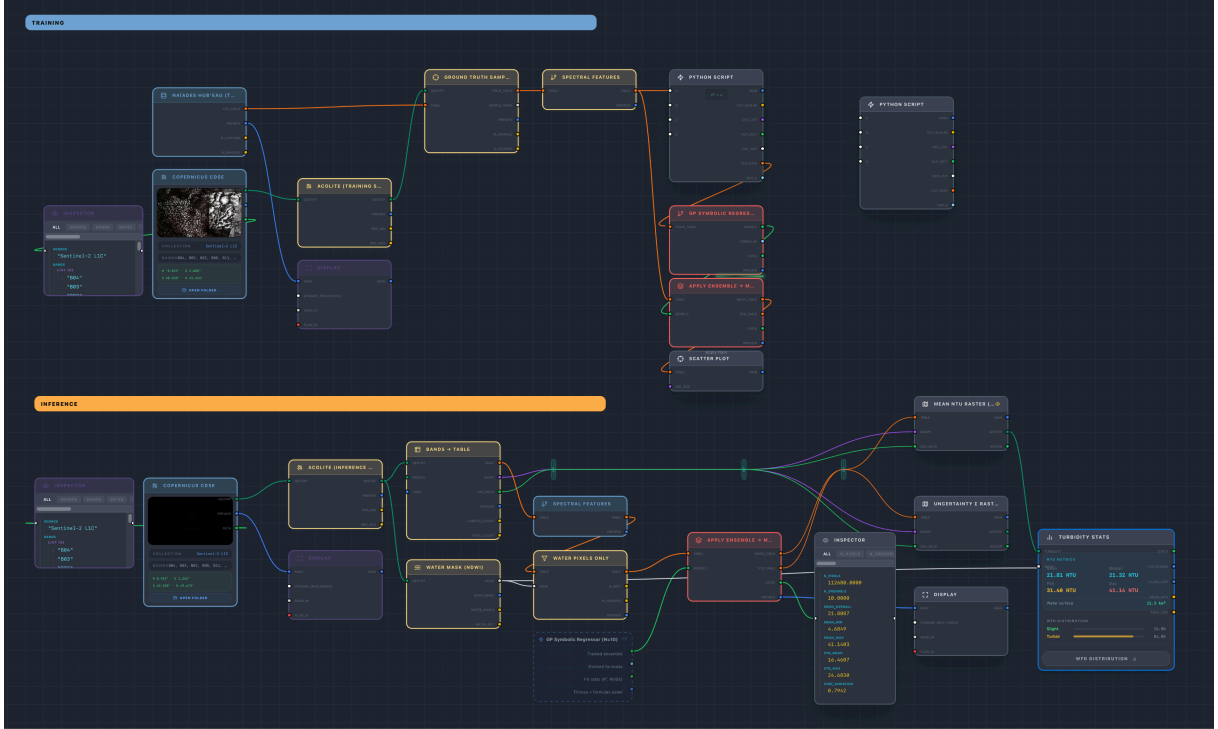


Figure 2: Actual VNStudio canvas screenshot. Top swimlane: training branch (Naïades download $\rightarrow$ ACOLITE $\rightarrow$ Ground-Truth Sampler $\rightarrow$ Spectral Features $\rightarrow$ logic_python rebalancer $\rightarrow$ GP Ensemble). Bottom swimlane: inference branch (Copernicus $\rightarrow$ ACOLITE $\rightarrow$ NDWI mask $\rightarrow$ Bands-to-Table $\rightarrow$ Apply Ensemble $\rightarrow$ Mean/Std rasters $\rightarrow$ Turbidity Stats). The entire experiment state lives in the JSON serialization of this canvas.

3.5 Symbolic regression ensemble (ml_symbolic_regressor)

We run `gplearn` 10 times with seeds $\{423, 424, \dots, 432\}$. Common parameters:

Parameter	Value
<code>target_column</code>	<code>label</code>
<code>n_ensemble</code>	10
<code>population_size</code>	1000
<code>generations</code>	50
<code>parsimony</code>	0.001
<code>test_size</code>	0.20
<code>log_transform</code>	True
<code>use_log / use_sqrt / use_div</code>	True / True / True

The function set is $\{+, -, \times, \div_{\text{protected}}, \sqrt{}, \log\}$. Training is performed in $\log(1 + y)$ space, inverted at inference. Label distribution justifies the log transform: mean 13.02 NTU, $\sigma = 25.20$, range $[0.12, 240]$.

3.6 Class-imbalance mitigation via stratified subsampling

A direct GP run on the raw 563 matchups collapsed the ensemble to a near-constant predictor (predicted-vs-observed slope ≈ 0.02): the parsimony penalty plus the heavy concentration of stations in the *Clear* class (NTU < 5 , $\approx 80\%$ of training rows) makes a constant intercept a locally optimal expression. We therefore insert a `logic_python` node in the training branch that retains only 30% of the *Clear*-class rows (random subsample, seed 42) while keeping all rows with NTU ≥ 5 :

```

clear_mask      = df['label'] < 5.0
df_clear_sub    = df[clear_mask].sample(frac=0.30, random_state=42)
df_turbid       = df[~clear_mask]
df_bal          = pd.concat([df_clear_sub, df_turbid], ignore_index=True)

```

This reduces the training set to ≈ 248 rows but shifts the predicted-vs-observed slope from 0.02 to **0.26**, the regime reported in §4. The choice of 30% is empirical; sweeping this value is left as a sensitivity analysis (§7).

3.7 Water mask (`geo_ndwi_mask`)

NDWI = $(G - NIR) / (G + NIR)$, thresholded at 0.1 and eroded by 2 px to remove mixed shoreline pixels.

3.8 Inference and raster reconstruction

The masked inference scene is unfolded to a per-pixel table (`geo_bands_to_table`), passed through the trained ensemble (`ml_ensemble_apply`, `clip_min=0`, `clip_max=50`), then folded back to 2-D rasters using the original pixel indices (`geo_table_to_raster`). Both μ and σ maps inherit the CRS and affine transform of the inference scene.

3.9 Statistics (`geo_turbidity_stats`)

WFD-style turbidity classes (*Crystal*, *Clear*, *Slightly turbid*, *Turbid*, *Very turbid*, *Extremely turbid*) are computed over the water mask. Parameters: `pixel_m = 10`, `max_ntu = 200`, `hist_max = 100`.

4 Results

4.1 Training set

After the spatial and temporal joins, **563 satellite–station matchups** were assembled. Following the stratified subsampling of §3.6 (retaining 30% of NTU < 5 stations), ≈ 248 rows entered the GP search. Table 1 summarizes the input feature distribution on the full 563-row pool, before rebalancing; reflectances after DOS-1 lie within physically plausible bounds.

Table 1: Training-set descriptive statistics (n = 563).

Feature	Mean	Std	Min	Max
Blue	0.054	0.057	0.002	0.200
Green	0.049	0.052	0.003	0.200
Red	0.070	0.051	0.026	0.200
NIR	0.101	0.049	0.004	0.200
SWIR1	0.080	0.048	0.002	0.200
SWIR2	0.056	0.043	0.001	0.193
label (NTU)	13.02	25.20	0.12	240.00

4.2 Best-of-ensemble model

The best individual across the 10 GP runs converges on the compact expression

$$\text{NTU} \approx \exp\left(\underbrace{\frac{\text{Blue}}{\text{Red}}}_{\text{clarity ratio}} + \underbrace{\frac{\text{NIR}}{\text{SWIR1}}}_{\text{SPM proxy}} - \underbrace{\text{Green}}_{\text{baseline}} \right). \quad (1)$$

(The exp wrapper inverts the log-target.) The three terms admit physical readings: Blue/Red is the inverse Forel–Ule clarity index; NIR/SWIR1 captures residual reflectance in spectral regions where pure water absorbs strongly, so any signal indicates suspended particulate matter; the $-$ Green term acts as a scene-brightness baseline. That this combination emerges from an unconstrained search — without bands being labelled by their hydrological role — is a tangible reproducibility check against the structural skeleton of the Nechad and Dogliotti models.

4.3 Inference over the Rouen reach

After NDWI masking, 112680 valid water pixels ($\approx 11.3\text{ km}^2$) entered inference. Ensemble statistics on the μ -map:

Table 2: Ensemble mean (μ) statistics over the inference scene, before vs after the stratified-subsampling rebalancing of §3.6.

Metric	Baseline (NTU)	Rebalanced (NTU)
Mean	5.07	21.81
Median	3.53	21.32
P90	6.21	31.40
Maximum	45.27	41.14

Ensemble spread (σ) and dispersion:

Metric	Value
Mean σ	3.78 NTU
Maximum σ	21.41 NTU
Coefficient of variation σ/μ	0.87

WFD class distribution by water area (Table 3) confirms a low-flow summer regime: 81 % of pixels classify as *Clear* (1–5 NTU), with no *Very turbid* or *Extremely turbid* water.

Table 3: WFD-class proportions over the 11.3 km^2 masked water surface.

Class	NTU range	% surface
Crystal	0–1	4.7
Clear	1–5	80.9
Slightly turbid	5–15	9.5
Turbid	15–50	5.0
Very turbid	50–200	0.0
Extremely turbid	>200	0.0

4.4 Figures

Training-set fit. A predicted-vs-observed regression on the rebalanced training set ($n \approx 248$, see §3.6) yields a positive but shallow slope of **0.26** (intercept 7.17 NTU): the model captures the direction of the NTU gradient yet systematically underestimates the upper tail, with observations above 100 NTU predicted in the 30–65 NTU range. The baseline run without stratified subsampling produces a near-constant predictor (slope 0.02), so the rebalancing recovers most of the signal even if the residual variance remains large.

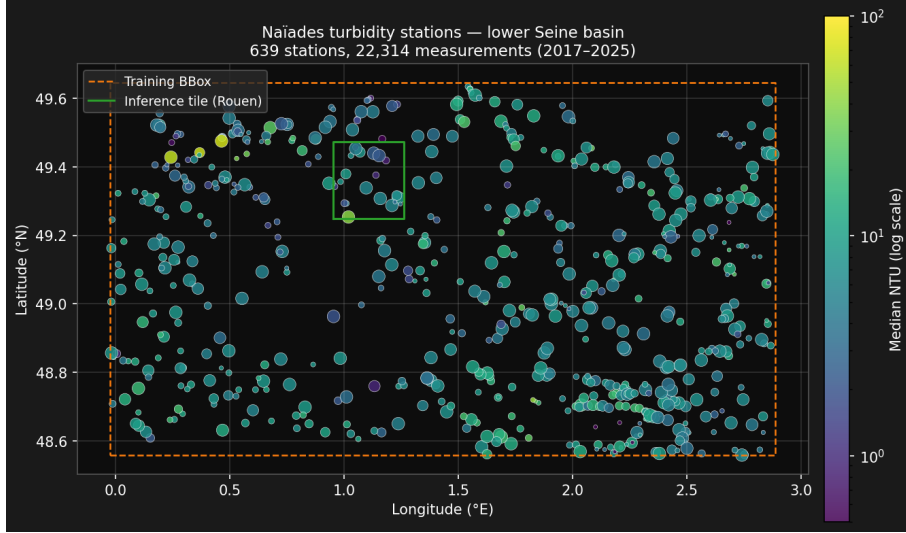


Figure 3: Naïades turbidity stations within the training bounding box (orange dashed, 2017–2025): 639 unique stations and 22,314 raw measurements, after filtering to $\text{NTU} \in [0, 500]$. Stations colored by median NTU (log scale); point size scales with the number of measurements. The green rectangle delineates the Rouen inference tile used in §4.

5 Discussion

5.1 Why ensemble matters

The coefficient of variation $\sigma/\mu \approx 0.87$ is the headline of the uncertainty analysis. Although the μ -map (Fig. 4) appears spatially crisp, the ten ensemble members disagree by an amount comparable to the signal. Two factors plausibly drive this:

1. **Residual training-set imbalance.** Even after the stratified subsampling of §3.6, the upper tail ($\text{NTU} > 50$) is represented by fewer than 10 stations. Models extrapolate differently in that regime, producing the high- σ pixels visible in Fig. 6 (right) and the systematic under-prediction of stations above 100 NTU reported in §4.
2. **Search-time stability.** GP rewards models that minimize training residuals but does not penalize input sensitivity. Adding a stability term across band perturbations would likely collapse the ensemble spread, at the cost of expressiveness.

In practice, the μ -map should be read as a class-level estimate (“Clear vs not-Clear”), and the σ -map should be consulted before any pixel-precise NTU claim. This is consistent with the proof-of-concept framing of §1.

5.2 Rediscovery of physical structure

Equation 1 is structurally close to Nechad’s red-band formula and Dogliotti’s blended Red/NIR estimator: both rely on a clarity-sensitive visible ratio and an SPM-sensitive infrared term. That the GP search recovers this structure without supervision is, at minimum, a sanity check on the data preparation chain. It also suggests that fixed bio-optical models occupy a local optimum that purely empirical methods naturally drift toward when given enough freedom.

5.3 Visual programming as a reproducibility primitive

A common failure mode in remote-sensing experimentation is undocumented preprocessing: a Sentinel-2 scene is loaded, “cleaned”, “deginted”, “re-projected”, and by the time the regression

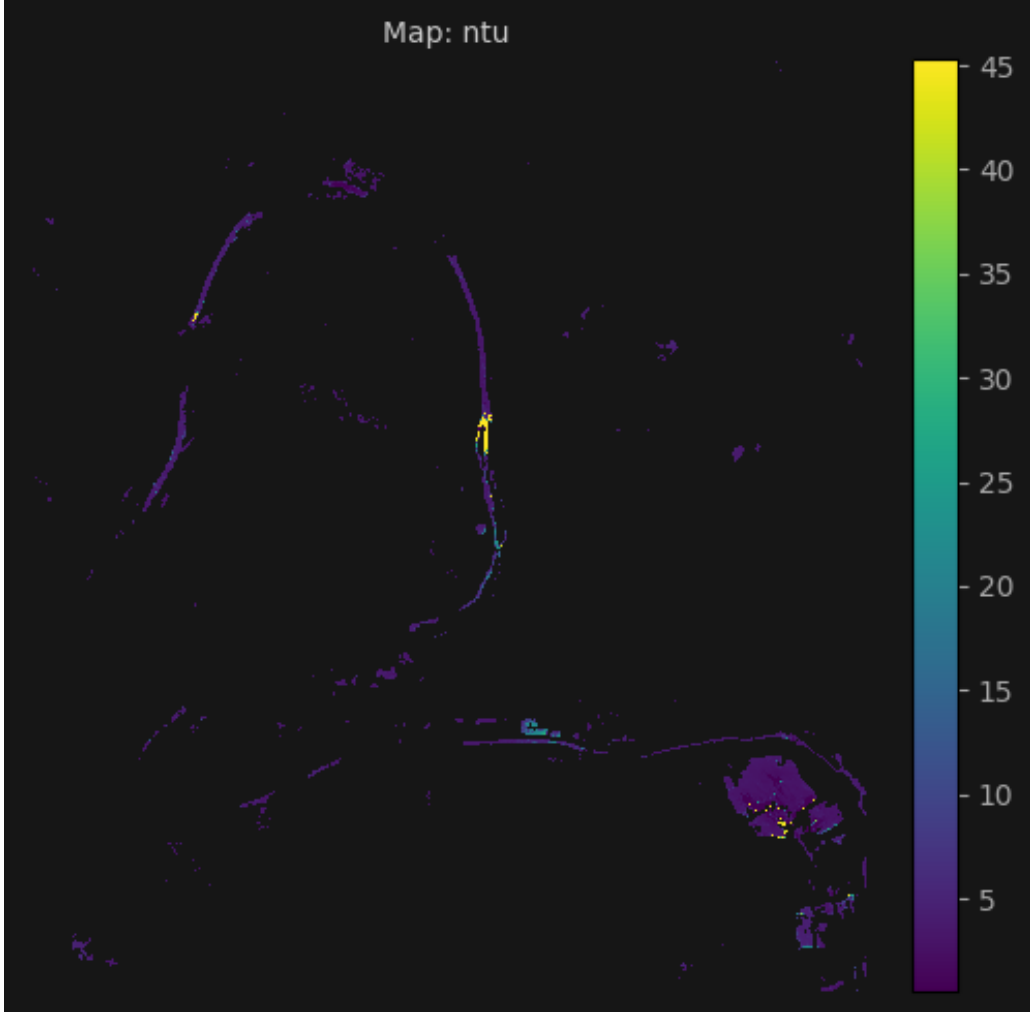


Figure 4: Ensemble mean turbidity (μ) over the Rouen reach (10 m, 2023-07-04). Color scale: 0–50 NTU, NDWI water mask applied. The navigable channel is uniformly *Clear* (1–5 NTU); turbid pockets concentrate along shorelines and side-channels.

is run the chain has accumulated ten implicit decisions that the manuscript does not record. A node-graph substrate forces every decision into a connector on a node, every parameter into a serializable field, and the whole graph into a single JSON file that can be re-opened and re-executed. We argue that this is not a UI niceness but a methodological prerequisite for the kind of multi-step, multi-source workflows that turbidity remote sensing entails.

6 Limitations

- DOS-1 is a fast approximation of atmospheric correction. For publication-grade Rrs the `geo_acolite_full` node also exposes the full ACOLITE DSF endpoint (`.SAFE` input), at ≈ 60 s per scene.
- Naiades stations are biased toward monitored rivers; the model extrapolates poorly to estuarine plumes lying outside the training footprint.
- The ± 30 d temporal window is generous. Tightening to ± 7 d drops training to ≈ 100 matchups but removes seasonal aliasing.
- GP *best-of-ensemble* reporting hides multimodality: more than one expression family typ-

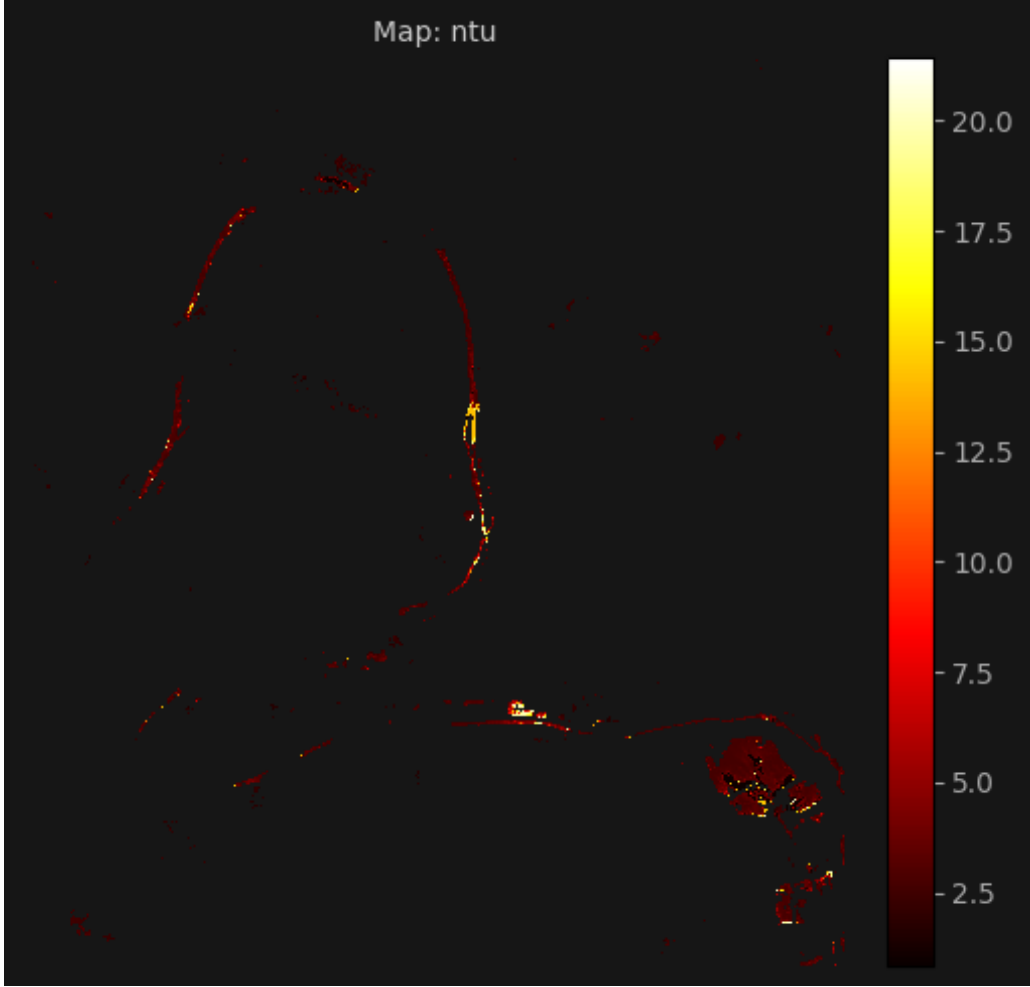


Figure 5: Ensemble standard deviation (σ) over the same scene. High- σ pixels concentrate at land/water boundaries and near morphologically complex side-channels — regions where individual GP models extrapolate differently.

ically reaches near-equivalent test RMSE. Reporting only one obscures this.

- The predicted-vs-observed slope of 0.26 (§4) shows the model captures direction but underestimates magnitude in the upper tail. Predictions saturate around 40–65 NTU while observations extend to 240 NTU.

7 Future work

The proof-of-concept invites several concrete extensions, ordered roughly by expected effort:

Sample weighting in place of subsampling. The current rebalancing (§3.6) discards $\approx 70\%$ of the Clear-class rows. Passing a `sample_weight` vector to `gplearn.SymbolicRegressor` (inverse class frequency, or a smooth density-based scheme) would preserve all training points and likely tighten the upper-tail predictions. This requires a small patch on `ml_symbolic_regressor` to accept a `weight_col` parameter.

Tighter temporal matchups. The current ± 30 d window introduces seasonal aliasing. A second pass at ± 7 d on a multi-year, multi-tile dataset would test whether the underestimation is genuine model bias or a temporal mismatch.

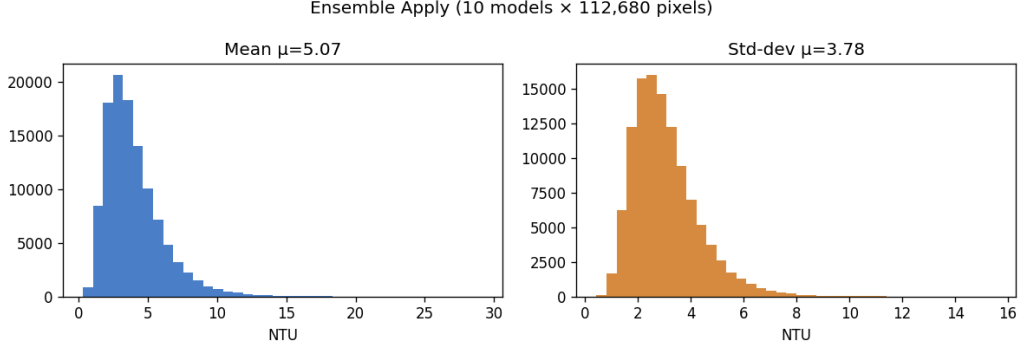


Figure 6: Pixel-wise distribution of μ (left) and σ (right) across the 112,680 water pixels of the inference scene.

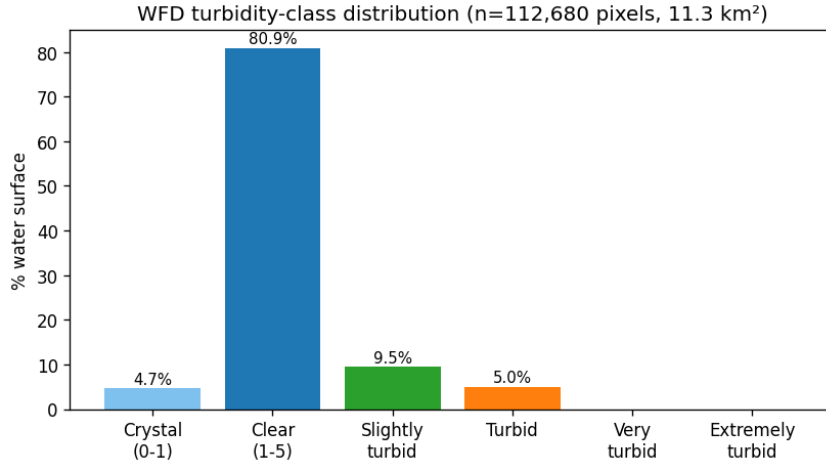


Figure 7: WFD turbidity-class distribution: 81% of the masked water surface is *Clear* (1–5 NTU).

Calibrated uncertainty. The ensemble σ is uncalibrated — there is no guarantee that $[\mu - 2\sigma, \mu + 2\sigma]$ covers 95% of the true distribution. A standard conformal-prediction wrapper [1], trained on the held-out 20% test split, would produce coverage-guaranteed intervals without retraining the GP.

Spatial transfer. The Rouen inference tile lies within the training footprint. Applying the model to a held-out basin (Loire, Rhône) would test whether the recovered symbolic structure generalises or whether basin-specific atmospheric / glint conditions drive the fit. The graph supports this directly: swap the `geo_copernicus` BBOX and re-run inference, leaving the trained GP ensemble untouched.

Full ACOLITE DSF. The DOS-1 fast path used here approximates atmospheric correction. Switching the `geo_acolite_full` node to its `.SAFE` mode (60 s per scene) would provide publication-grade Rrs and remove a confound from the comparison with published bio-optical models.

Multi-target GP. The same pipeline structure applies unchanged to chlorophyll-a, DOC, or total suspended solids, all available through the Hub'Eau API. A single graph could be parametrized over target column and bio-optical preset, providing a directly comparable benchmark suite across water-quality variables.

Expression-family analysis. Reporting only the best-of-ensemble formula hides the family of structurally distinct expressions that reach near-equivalent test RMSE. A clustering analysis of evolved formulas (by AST distance or by output-space similarity) would expose this multi-modality and may surface alternative physical interpretations.

8 Reproduction recipe

1. Clone VNStudio:
`git clone https://github.com/Nikos-Unilasalle/VisionNodes && cd VisionNodes`
2. Install:
`npm install && pip install -r engine/requirements.txt`
3. Launch:
`npm run studio`
4. Open the template:
`public/templates/Machine Learning Course/gp_ensemble_uncertainty.vn`
5. Click *Fetch* on the **Naiades** node (live Hub'Eau API).
6. Click *Fetch* on both **Copernicus** nodes (needs Copernicus credentials in `~/.config/copernicus.json`).
7. Click *Run* on both **ACOLITE** nodes (<5s each, DOS-1 mode).
8. Click *Train* on the **GP Ensemble** node ($\approx 30\text{--}60\text{ s}$ for $10 \times 50 \times 1000$ GP search on a modern laptop).
9. The μ / σ rasters and the Turbidity Stats panel update automatically.

Data and code availability

The complete pipeline (graph, plugins, and template) is available at <https://github.com/Nikos-Unilasalle/VisionNodes>. The training table and inference rasters used in this manuscript are deposited at [Zenodo DOI placeholder].

Acknowledgements

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